

- Q.28 Describe how SFRs are accessed and give examples of commonly used SFRs in 8051.
- Q.29 Explain PUSH and POP instruction.
- Q.30 What are the advantages of SCADA over other protocols?
- Q.31 Explain LCD interfacing with 8051.
- Q.32 List of building blocks of PLC and write their functions.
- Q.33 What is mask? Explain its function in Masked Equal to (MEQ) instruction.
- Q.34 Write a short note on "Counters of 8051 Micro controller".
- Q.35 Explain working of motor on-off operation using ladder diagram programming.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write short note any two:
- Comparison instruction of PLCs
 - Application of PLCs in bottling operations.
 - Interfacing keypad with 8051 micro controllers
- Q.37 Explain different Boolean instruction of PLC with the help of ladder diagram.
- Q.38 Write programs to fill external RAM location from 0100H to 0400H with data 65H.

No. of Printed Pages : 4 120962/105864/031065A
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**6th Sem / Electrical Engg., Eltx, Mecatronics, GE,
Power Station Engg., Elect & Eltx. Engg.
Sub : Programmable Logic Controller & Microcontroller
/ Mic. Cont. & PLCs**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 No. of interrupt sources in 8051 are ____.
- Five
 - Seven
 - Four
 - Six
- Q.2 OTL ____ instruction.
- Output
 - Input
 - Memory
 - None
- Q.3 ____ instruction will clear a memory location by putting a zero when the input to the function is true.
- CR
 - CLR
 - CLRE
 - CLL
- Q.4 Which of the following is not a PLC Programming language.
- SFC
 - LD
 - IL
 - CTU
- Q.5 Which addressing mode is used in instruction LXIH, 4500H.
- Direct
 - Indirect
 - Register
 - Implicit

- Q.6 An AND function implemented in ladder logic uses
- Normally closed contacts in series
 - Normally open contacts in series
 - A single normally closed contact
 - Normally open contacts in parallel
- Q.7 Ladder logic has been developed to mimic ____.
- Drives
 - Instruction
 - Relay
 - Register
- Q.8 Which is the bit size of 8051 microcontroller?
- 4bit
 - 8 bit
 - 16 bit
 - 32 bit
- Q.9 Which of the following is a type of microcontroller interrupt?
- INT0
 - INT1
 - TFQ
 - All of the above
- Q.10 The CPU of a PLC is responsible for
- Providing power to the input and output modules
 - Communicating with other devices on a network
 - Executing the control program and making decisions
 - Interfacing with human operators.

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 8051 microcontroller is a _____ Pin IC.

(2) 120962/105864/031065A

- Q.12 FBD stands for ____.
- Q.13 What is the clock frequency range of the 8051 microcontrollers?
- Q.14 Give the name of any two PLC manufacturer?
- Q.15 What is Relay?
- Q.16 What is MOV instruction?
- Q.17 What is compiler?
- Q.18 What is the function of SCON instruction?
- Q.19 Draw the symbol of XIC.
- Q.20 How many bank memory banks are used in internal memory of 8051?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the implementation of OR gate using a ladder diagram.
- Q.22 Explain the role of Port 0 in the 8051 and its dual functionality.
- Q.23 What is Ladder programming? Draw its components.
- Q.24 Write the basic features of PIC microcontrollers.
- Q.25 Differentiate the function of Assembler, Compiler and Interpreter.
- Q.26 Discuss the procedure of interfacing stepper motor with microcontroller.
- Q.27 Draw the Pin Diagram of 8051 microcontroller.

(3) 120962/105864/031065A